

Lecture 2

$$\frac{1}{\ell} \sum_{i=1}^{\ell} [\underbrace{a(x_i)}_{y_i} \neq y_i] \rightarrow \min_a$$

$$y_i \in \{1, \dots, c\}$$

$$p = \begin{pmatrix} p(\hat{y}_i = 1) \\ p(\hat{y}_i = 2) \\ \vdots \\ p(\hat{y}_i = c) \end{pmatrix}$$

$$z_i = \sigma(W_i z_{i-1} + b_i)$$

$$z_n = W_n z_{n-1} + b_n$$

$$z_n \in \mathbb{R}^c$$

$$W_n \in \mathbb{R}^{d_{n-1} \times c}$$

$$p = \{p_i\}_{i=1}^c, p_i \geq 0, \sum_{i=1}^c p_i = 1$$

$$z_n = (z_1, \dots, z_c)$$

$$p_i = \frac{\exp(z_i)}{\sum_{j=1}^c \exp(z_j)} \geq 0$$

softmax

z - logits

p - probabilities

$$z_j > z_i \Rightarrow p_j > p_i$$

$$y\text{-target} \in \{1, \dots, c\}$$

$$\underbrace{- \sum_{k=1}^c [k=y] \log p_k}_{-\log p_y''} \rightarrow \min_{(W, b)}$$

Negative Log Likelihood (NLL)

$$- \sum \log p_{y_i} \xrightarrow{\updownarrow} \min$$

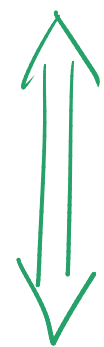
$$\prod p_{y_i} \rightarrow \max$$

$$\{(x_i, y_i)\}_{i=1}^{\ell} \text{ — training set}$$

$$-\sum_{i=1}^l \sum_{k=1}^c [k=y_i] \log p_k^{(i)} \rightarrow \min_{W,b}$$

$$\parallel$$

$$\log p_{y_i}^{(i)}$$



$$-\sum_{i=1}^l \log p_{y_i}^{(i)} \rightarrow \min \iff \prod p_{y_i}^{(i)} \rightarrow \max$$

$$NLLLoss(\text{Softmax}(z), y) \rightarrow \min$$

$$\log \text{softmax}(z_i) = \log \frac{\exp(z_i)}{\sum_{j=1}^c \exp(z_j)} =$$

$$= \underbrace{\log \exp(z_i)}_{\parallel z_i} - \underbrace{\log \left(\sum_{j=1}^c \exp(z_j) \right)}_{\log \text{sumexp}(z)}$$

$$\log \frac{\exp(z_i - \max_j z_j)}{\sum_{j=1}^c \exp(z_j - \max_k z_k)}$$

CrossEntropyLoss ($\underset{\substack{\uparrow \\ \text{logits}}}{z}, y$)

torch.softmax

$$CE(\{p_i\}_{i=1}^c, \{q_i\}_{i=1}^c) = - \sum_{i=1}^c q_i \log p_i =$$

$$= - c \frac{1}{c} \log \frac{1}{c} = - \log \frac{1}{c} \quad \nearrow$$

$$q_i = [i=y]$$

$$\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad y \quad \begin{pmatrix} p_1 \\ \vdots \\ p_c \end{pmatrix}$$

$$\frac{1}{10c} \quad \frac{1}{10c} \quad \dots \quad \frac{y_c-1}{10c} \quad \dots \quad \frac{1}{10c}$$

label smoothing
blending (mixup)

$$\alpha V_1 + (1-\alpha) V_2$$

$$y_1 \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad y_2 \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

$$V_1 \quad V_2$$

Dropout

$$z_k \in \mathbb{R}^{B \times d_k}$$

B - batch size

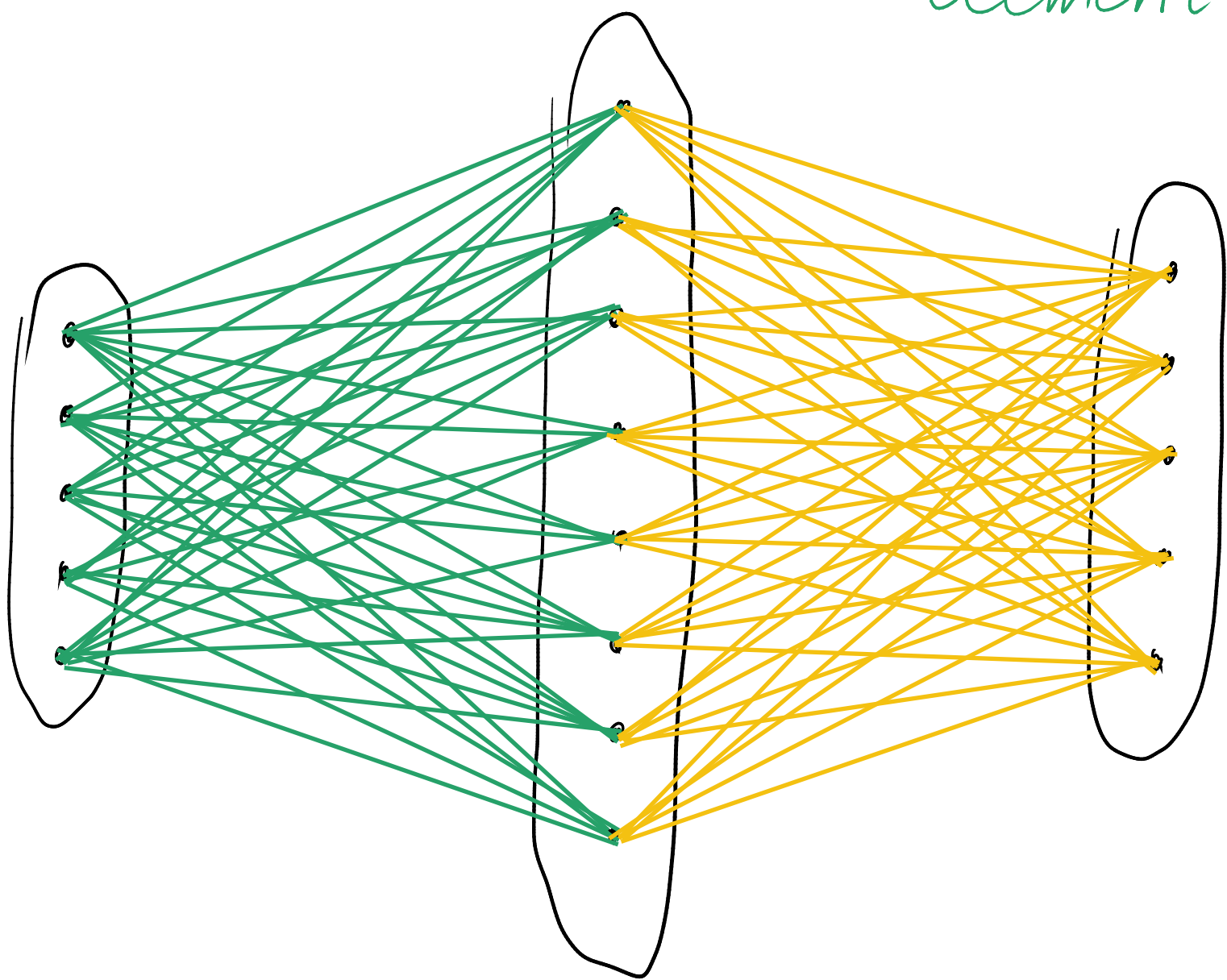
d_k - hidden size

$$m \in \{0, 1\}^{B \times d_k}$$

$$m_{ij} \sim \text{Bern}(1-p)$$

$$y = \text{dropout}(z_k) = m \odot z_k$$

element-wise product



$$\begin{aligned} \mathbb{E}[y_i] &= \mathbb{E}[m_i \odot z_k^{(i)}] = \mathbb{E}[m_i] \odot z_k^{(i)} = \\ &= (1-p) z_k^{(i)} \end{aligned}$$

Train mode:

$$m_{ij} \sim \text{Bern}(1-p)$$

$$y = m \odot z_k \frac{1}{1-\rho}$$

$$\mathbb{E}[y_i] = z_k^{(i)}$$

Eval mode:

$$y = z$$

model.eval()

model.train()

no gradient changes!!!

Batch Normalization

(0, 255)



$$|z_1^{(i)}| \sim 1$$

$$\alpha < 1$$

$$|z_k^{(i)}| \sim 1 \cdot \alpha^k$$

$$\alpha > 1$$

$$z_k^{(i)} \rightarrow \frac{z_k^{(i)}}{\sum_j |z_k^{(j)}|} \sim 1$$

$$\mathbb{E} z_k^{(i)} = 0$$

$$\mathbb{D} z_k^{(i)} \sim 1$$

Train mode

$$Z \in \mathbb{R}^{B \times d_k}$$

$$Z_i \in \mathbb{R}^{d_k}$$

$$\mu = \frac{1}{B} \sum_{i=1}^B Z_i$$

$$\sigma^2 = \frac{1}{B} \sum_{i=1}^B (Z_i - \mu)^2 \quad \mu, \sigma^2 \in \mathbb{R}^{d_k}$$

$$\hat{Z}_i = \frac{Z_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

$$\frac{1}{B} \sum_i \hat{Z}_i = 0$$

$$\frac{1}{B} \sum \hat{Z}_i^2 = 1$$

$$y_i = \hat{Z}_i \odot \omega + b$$

$$\omega, b \in \mathbb{R}^{d_k}$$

↑
learnable

$$\frac{1}{B} \sum y_i = b$$

$$\frac{1}{B} \sum (y_i - b)^2 = \omega^2$$

μ, σ - collect

$$\text{running-mean}_m := \text{running-mean}_{m-1} \cdot$$

$$\cdot (1-\gamma) + \mu_m \cdot \gamma \quad \frac{B}{B-1} \quad \gamma \approx 0.1$$

$$\text{running-var}_m := \text{running-var}_{m-1} (1-\gamma) + \sigma_m^2 \cdot \gamma$$

Eval mode

$$\hat{z}_i = \frac{z_i - \text{running-mean}}{\sqrt{\text{running-var} + \epsilon}}$$

$$y_i = \hat{z}_i \odot \omega + b$$